Ansh Agrawal

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Professional Summary

Fourth-year B.Tech student specializing in AI/ML with strong foundations in software engineering and applied machine learning. Experienced in building scalable applications and data-driven solutions through academic projects and internships.

Education

B.Tech in Computer Science and Engineering (AI/ML) 2023 – 2027
UPES Dehradun CGPA: 8.09
Pragati Vidya Peeth, Gwalior | CBSE (Class XII) April 2023
St. John Vianney School, Gwalior | CBSE (Class X) April 2021

Technical Skills

Programming Languages: Java, C, Python, SQL
AI & Machine Learning: Scikit-learn, Pandas, NumPy, XGBoost, Logistic Regression, Random Forest, Reinforcement Learning, Feature Engineering, SHAP
Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Management Systems (DBMS)
Tools: Git, GitHub, Jupyter Notebook, VS Code

Projects

Agentic Recommendation System *Python, Pandas, Reinforcement Learning*

- Developed an agentic recommendation system using Item-Item Collaborative Filtering, ε-greedy Multi-Armed Bandit, and Q-Learning for session-aware product recommendations.
- Processed a 50K-event subset of the RetailRocket dataset by sessionizing user interactions and generating Top-K recommendations using cosine similarity.
- Improved Precision@5 and HitRate@5 by approximately 200% over the collaborative filtering baseline through session-aware reinforcement learning.

Water Quality Analysis *Python, Scikit-learn, Pandas*

- Built ML model on 100K+ samples (23 features) using Random Forest, achieving 88.2% accuracy and 97.3% recall
- Implemented preprocessing including imputation, MinMax scaling, one-hot encoding, and feature engineering
- Identified Manganese, pH, and Chloride as top predictors; validated with 0.94 ROC-AUC

Work Experience

AIML Intern June 2026 – July 2026
IBM SkillsBuild

- Built a Credit Risk Assessment System for underbanked applicants, translating XGBoost default-probability scores into approve/decline decisions via a validation-tuned threshold.
- Benchmarked against Logistic Regression and a dummy baseline on 307K+ applicant records, improving test ROC-AUC to 0.769 and lift to 1.90x over the base default rate to support lending decisions.
- Delivered SHAP-based explanations for decline decisions and shipped an interactive interface for non-technical stakeholders to review credit risk scores.

Sustainability Intern June 2025 – July 2025
Enviraj Consulting Pvt. Ltd.

- Designed GHG emission inventories for clients aligned with IPCC and DEFRA standards
- Built the frontend UI of a responsive real-time carbon calculator using HTML5 and Tailwind CSS
- Collaborated on GHG Protocol documentation, analyzing Scope 1-3 emissions for reduction strategies

Certifications

- AI Tools & ChatGPT Workshop — be10x July 2026